Interstitial Lung Disease: Foundations and Frameworks

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Adapted from: Interstitial Lung Disease: Foundations and Frameworks

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Explain the use of the term interstitial lung disease (ILD).
2. Describe the clinical symptoms for patients with interstitial lung disease.
3. List the most common causes of interstitial lung disease, including medications.
4. Explain how interstitial lung disease affects pulmonary pathology and physiology.
5. Summarize the diagnostic approach to ILD.

CASE CONNECTION

RM is a 62-year-old female presenting with a progressive dry cough and shortness of breath of 6 months' duration. "I just can't get through a set of tennis anymore," she says. "Before, I could play 2 sets without problems." One year ago, RM was diagnosed with Hodgkin lymphoma and treated with the standard chemotherapy regimen of ABVD (adriamycin, bleomycin, vinblastine, and dacarbazine). She is not currently taking any medications. She stopped smoking 1 year ago, after a 30-pack-year history. On exam, her vital signs are normal, with a respiratory rate of 16/minute. Lung auscultation reveals crackles one third of the way up the lung fields. Her pulse ox on room air is 92%.

What is the likely diagnosis? What would you expect the chest x-ray to show? What other testing would be helpful? Consider your answers as you read, and we'll revisit RM at the end of the brick.

GO TO CONCLUSION ↓

What is Interstitial Lung Disease (LO #1)?

Interstitial lung disease (ILD) is not a distinct disease, but rather an umbrella term encompassing more than 200 different pulmonary disorders that may present similarly from a clinical, pathologic, and imaging point of view. Its causes range from dust inhalation, to exposure to drugs and radiation, to idiopathic and genetic causes. ILD is characterized by the formation of fibrotic scar tissue within the lungs as a result of chronic inflammation. It causes restrictive lung disease, as the lung tissue gradually loses compliance and can't fully expand and contract with each breath.

Management generally focuses on symptom control and slowing progression of the underlying disease but does little to reverse existing fibrosis.

Fibrosis-related restrictive lung disease results from ILD.

How Do Patients with Interstitial Lung Disease Present (LO #2)?

Patients with ILD most often present with a persistent dry cough or difficulty breathing. Symptoms usually appear insidiously, so patients may not know the precise moment when they began. Lung fibrosis is progressive and irreversible without treatment. Diminished lung function can limit patients' everyday activities and quality of life.

Patients with ILD caused by systemic disorders or hypersensitivity may also experience constitutional symptoms like fever, chills or myalgia, rashes, or arthralgias. A history of occupational exposure to coal or silica, smoking (currently or in the past), or allergies may be reported. Current and previous medication, as well as any previous radiation (eg, as part of a cancer treatment protocol), should be documented, since some medications cause ILD.

Persistent dry cough or difficulty breathing are the key symptoms of ILD.

What Causes Interstitial Lung Disease (LO #3)?

ILD can be primary (idiopathic) or due to secondary causes.

Primary Interstitial Lung Disease

Idiopathic pulmonary fibrosis (IPF) is a type of lung fibrosis without an obvious secondary cause. Although by definition IPF is idiopathic (there is no known cause), some genetic or environmental factors may increase a patient's risk of developing it.

CLINICAL CORRELATION

Factors that increase a patient's risk of developing IPF include smoking, pollutants, viral infection, and chronic gastrointestinal aspiration.

Secondary Interstitial Lung Disease

Secondary ILD has an underlying identifiable cause. Secondary causes include:

- Drugs and radiation
- Infections
- Pneumoconiosis (inhaled dusts and toxins)
- Systemic inflammatory disorders

Drugs and Radiation. Several drugs cause ILD either through cytotoxicity or an immune-mediated mechanisms. Drug-induced ILD can develop during or shortly after treatment, usually in less than 1 year. Up to 40% of cases are asymptomatic and diagnosed by imaging.

Chemotherapy drugs—particularly **bleomycin**—cause drug-induced ILD. Other chemotherapy offenders include gemcitabine, used in lung and breast cancer; or immune checkpoint inhibitors (eg, pembrolizumab, ipilimumab).

Other drugs that may cause ILD include immunomodulating agents (eg, methotrexate, leflunomide), nonsteroidal anti-inflammatory drugs, some antibiotics (eg, nitrofurantoin), and antiarrhythmics (eg, amiodarone).

ILD can also be secondary to chest radiation, which is sometimes used to treat lung cancer or mediastinal tumors.

Infections. Many pneumonias, including viral, bacterial (including tuberculosis), and fungal, can be complicated by severe persistent inflammation leading to secondary ILD.

Pneumoconiosis. Some farming and industrial jobs expose workers to inhaled dust and toxins, such as asbestos, silica, and beryllium. They can result in secondary ILD.

Systemic Inflammatory Disease. Inflammatory disorders like rheumatoid arthritis, lupus, progressive systemic sclerosis, and polymyositis can cause secondary ILD as part of a systemic inflammatory response. Some involve an autoimmune response of the immune system against normal tissue. Sarcoidosis is characterized by the formation of granulomas in multiple organs, especially the lung. Most of these disorders are covered in the immunology collection.

Chemotherapy drugs, particularly bleomycin, are most likely to cause ILD.

What is the Pathophysiology of Interstitial Lung Disease (LO #4)?

The pathophysiologic mechanism of ILD varies depending on the underlying cause. However, some pathologic features are shared by all ILD, regardless of cause. In ILD, the pulmonary interstitium develops persistent **inflammation with excessive fibrosis** (scarring) along the small respiratory airways.

Inflammation and Fibrosis

What causes the inflammation? Persistent lung inflammation can be caused by systemic inflammatory conditions or immune hypersensitivity reactions. Toxic agents produce repetitive microinjuries along bronchioles and alveoli, damaging the respiratory epithelium and underlying basement membrane. Cells involved in immune defense, such as lymphocytes, neutrophils, and eosinophils, react quickly and signal local macrophages and endothelial cells to limit the destruction. The inflammation is generally mild enough to heal, but it does so imperfectly, with excessive fibrosis.

As a result of this fibrosis, the previously elastic lung tissue is replaced with thicker collagen fibers, limiting lung expansion and gas exchanges. This is termed **restrictive lung disease**, in which the thickened lung cannot expand

and fill with air during inspiration. There may also be impairment of gas exchange across the fibrosed alveolar membrane. The result is hypoxemia and dyspnea.

Changes in Lung Physiology

ILD is a restrictive disease marked by decreases in lung volumes. The composite lung capacities will also be affected, including decreases in the vital capacity, inspiratory capacity, functional residual capacity, and total lung capacity (TLC). These volumes and capacities are shown in Figure 1.

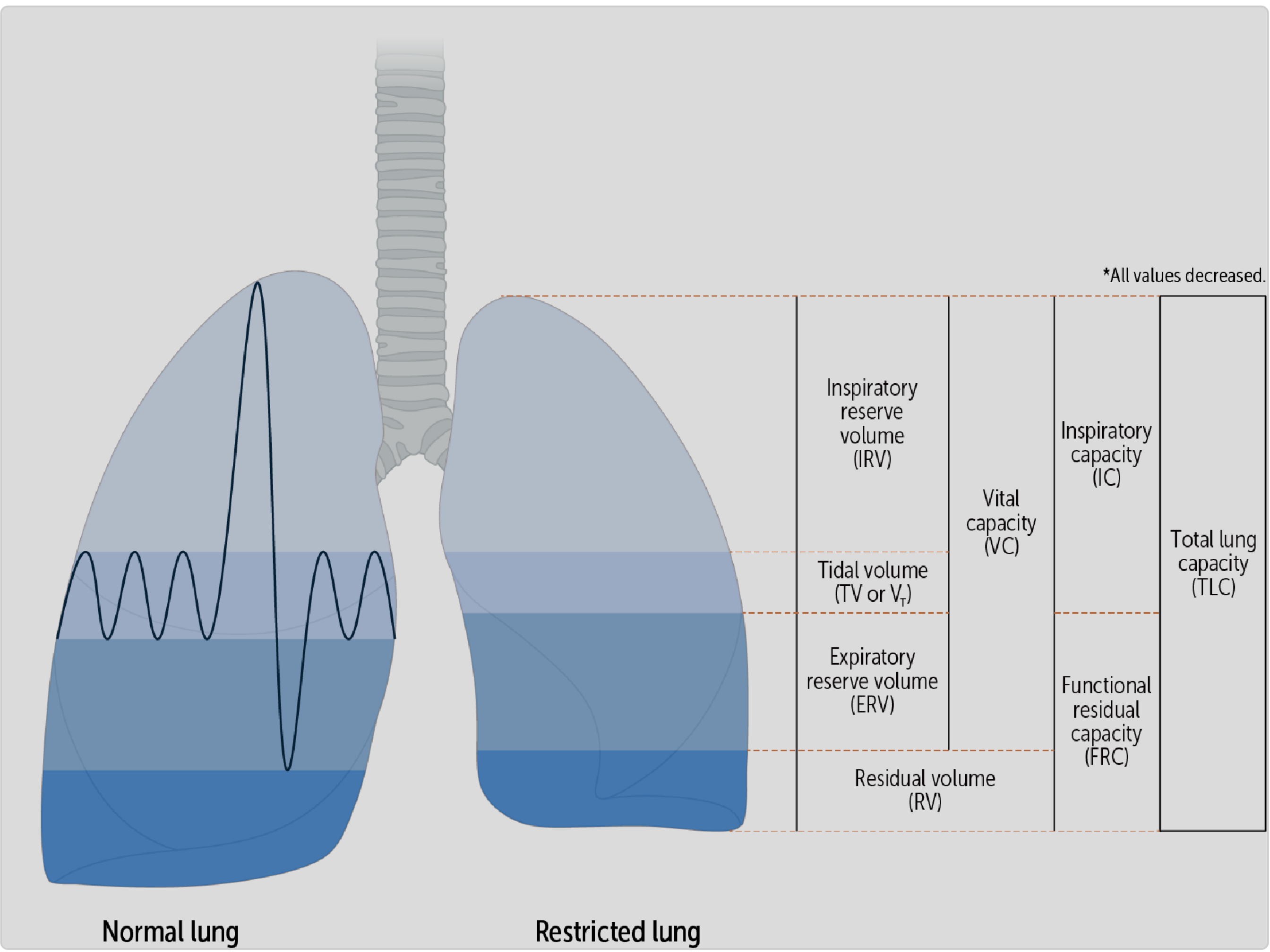

Figure 1 Lung volume changes in restrictive lung disease

The main pathophysiologic feature of ILD is fibrosis (scarring), caused by chronic inflammation within the lung tissue.

How Do We Diagnose Interstitial Lung Disease (LO #5)?

While the physical exam and history are often suggestive of ILD, imaging studies are the most useful diagnostic tool, supported by pulmonary function tests and blood tests.

Physical Examination

Lung auscultation in a patient with ILD often reveals abnormalities such as crackles or inspiratory rhonchi. Vital signs may show tachypnea (due to hypoxemia, seen on pulse oximetry).

In advanced stages, chronic hypoxia can cause cyanosis. Persistent peripheral tissue hypoxia leads to clubbed fingers (Figure 2).

Figure 2 Clubbed fingers in advanced lung disease

Additional findings may reflect underlying systemic disease, such as:

- Blue or painful fingers in response to cold (Raynaud phenomenon) in lupus and other connective tissue diseases.
- Cutaneous thickening in systemic sclerosis.
- Lymphadenopathy and erythema nodosum in sarcoidosis.
- Malar rash in lupus.
- Subcutaneous nodules in rheumatoid arthritis.

- Swollen, painful joints in rheumatoid arthritis.

Cyanosis results from decreased oxygenation of the peripheral tissues.

Imaging Studies

Radiologic imaging is the best initial test to diagnose ILD.

Chest X-Ray. A chest x-ray is often the first sign that a patient has ILD. However, up to 10% of chest radiographs in ILD may be normal.

When abnormal, the typical **chest x-ray** finding is a bibasilar reticular pattern (formation of multiple lines, forming a network appearance), as seen in Figure 3. However, nodular or mixed reticulonodular patterns can also be seen.

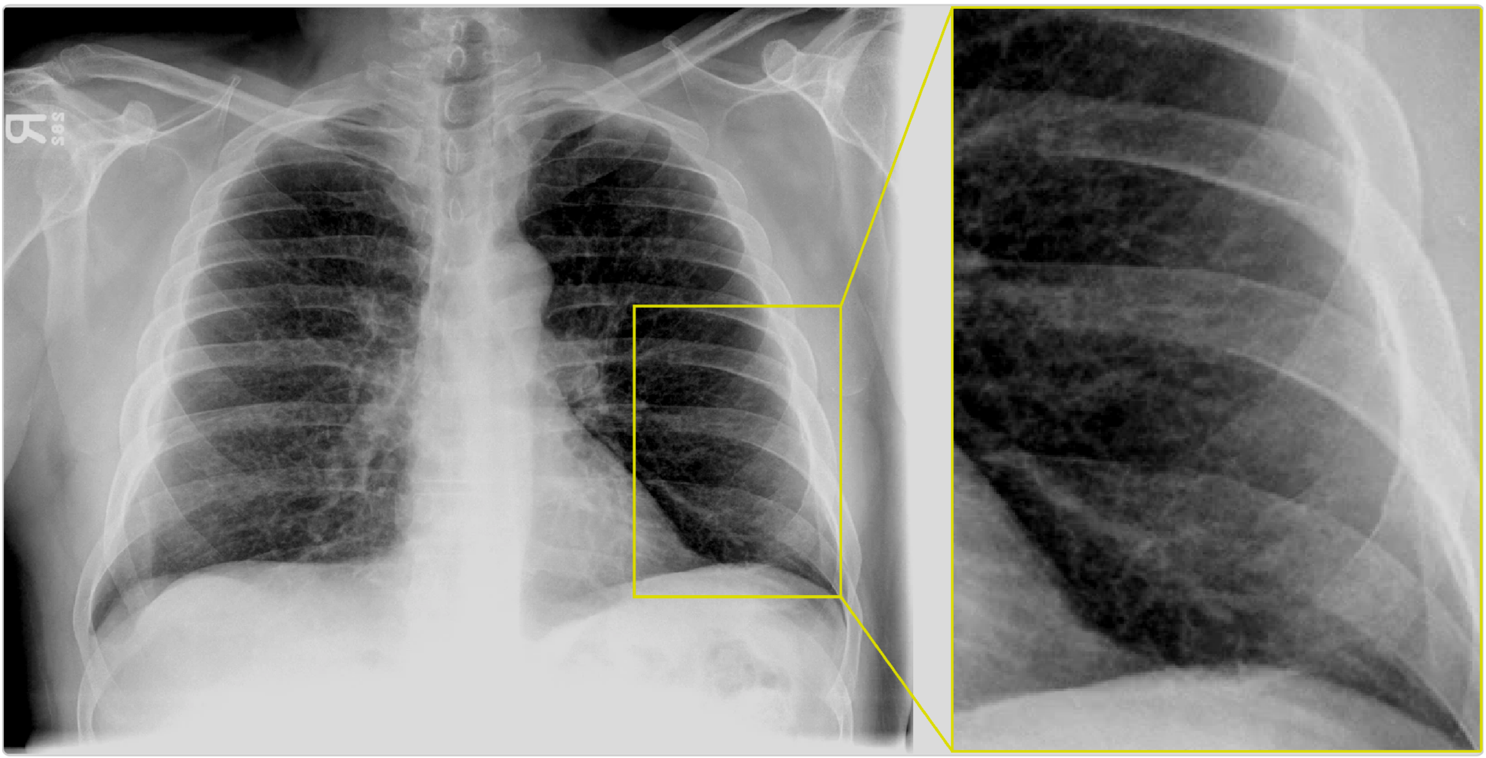

Figure 3 Chest x-ray showing a reticular pattern in ILD

In advanced ILD, a “honeycombing” appearance is seen (Figure 4). This is caused by small cystically dilated air spaces in the lung and is correlated with poor prognosis.

Figure 4 Chest x-ray showing honeycombing in ILD

High-Resolution CT Scanning. Computed tomography can show the extent and distribution of lung damage as well as features that may identify the likely cause of the ILD. The thick, well-defined walls of cystically dilated pulmonary air spaces often appear as a characteristic “honeycomb” pattern (as evidenced by the axial chest CT scan in Figure 5).

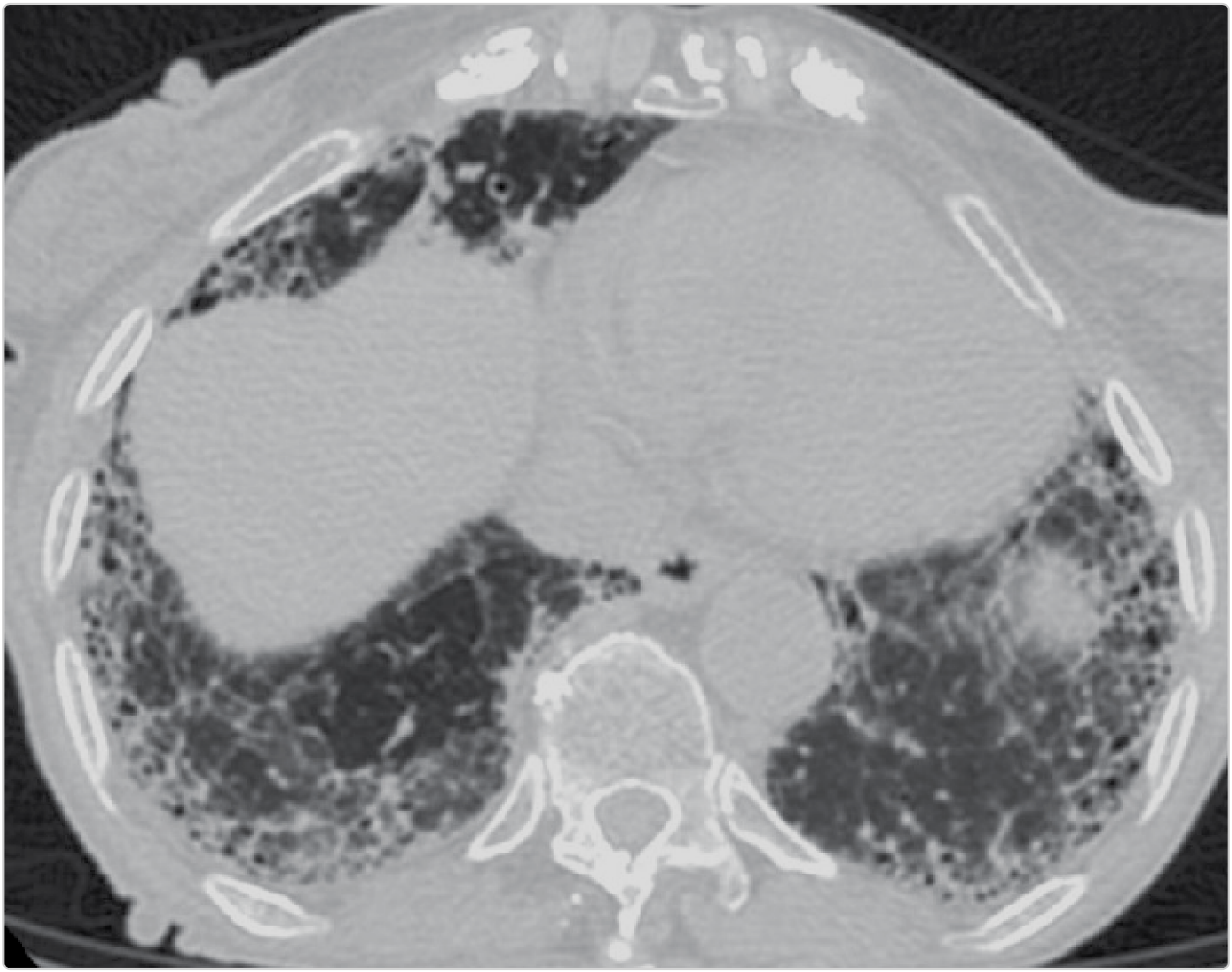

Figure 5 Axial CT of the chest showing honeycombing

The most typical finding on chest x-ray is a reticular pattern along both pulmonary bases.

Pulmonary Function Testing and Arterial Blood Gases

Pulmonary function testing will typically show a **restrictive pattern**. As a result, spirometry shows decreases in the forced vital capacity (FVC) and forced expiratory volume in 1 second (FEV_1). Since both decline proportionally, the FEV_1/FVC ratio remains normal, or slightly high, unlike the drop seen in obstructive diseases like asthma.

The diffusing capacity of the lung for carbon monoxide (DLCO) is also reduced. This is caused by a combination of alveolar and capillary fibrosis as well as ventilation/perfusion mismatch, because the dysfunctional parts of the lung tissue still maintain normal blood flow.

Arterial blood gas assessment may reveal moderate **hypoxemia**, which causes secondary hyperventilation and respiratory alkalosis (high pH and low partial pressure of carbon dioxide [PCO_2]). Respiratory acidosis (acidosis with high PCO_2) caused by extensive lung damage is much less common, only seen in patients with advanced stages of the disease.

CLINICAL CORRELATION

Respiratory alkalosis develops because the secondary hyperventilation causes the body to breathe off more CO_2. Less CO_2 causes a rise in the pH.

The restrictive changes are caused by the decreased compliance of the lung tissue from the progressive accumulation of collagen and consequent fibrosis.

Other Blood Tests

Blood and serum studies may identify clues regarding underlying cause. Hematologic (eg, eosinophilia, lymphocytosis), hepatic, or renal function abnormalities can guide further investigations. For a suspected autoimmune disorder, additional tests may include serologic studies, such as antinuclear antibodies and rheumatoid factor.

Antinuclear antibody and rheumatoid factor studies are serologic tests that can used to rule out a rheumatologic cause of ILD.

Tissue and Cell Examination

In some patients, a **bronchoalveolar lavage** (**BAL**) or lung biopsy is performed to determine the underlying cause of diagnosed ILD, as well as to exclude other lung diseases that may cause restrictive disease, such as some cancers.

BAL collects a sample of cells and fluid from distal bronchioles and alveoli. A bronchoscope is lowered into the targeted lung area, then sterile saline is injected into the airways to "pick up" some of the cell lining, which is then collected. The sample is analyzed to identify the cause. Additionally, BAL testing for hemorrhage, malignancy, or infection can help rule out other causes of patients' symptoms that on chest imaging can appear similar to ILD.

Lung biopsy can be performed endoscopically or via open surgical technique.

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to RM, what is the likely diagnosis? What would you expect the chest x-ray to show? What other testing would be helpful?

Based on her history and physical exam, you suspect that RM has interstitial lung disease as a result of her prior exposure to bleomycin, which was used to treat her Hodgkin disease. Chest x-ray reveals a bibasilar reticular and reticulonodular pattern. Pulmonary function tests reveal a restrictive pattern, with decreased FVC, an increased FEV_1/FVC ratio, and a reduction in the DLCO. "What's the next step?" RM asks. "I recommend that we consult a pulmonary specialist," you say. "Some new treatments are available that may be beneficial. I'd also like to get a high-resolution CT of the chest to evaluate the extent of the disease. That might help treatment decisions." "Sounds like a good idea," she says. "I'd really like to get back to the tennis court as soon as possible."

Summary

- Interstitial lung disease (ILD) includes multiple diseases that lead to lung interstitial fibrosis and restrictive lung disease.

- Common causes of ILD include exposure to environmental toxins, radiation, and systemic inflammatory conditions.
- Symptoms of ILD, such as dry cough and shortness of breath, may slowly progress over a long period of time.
- Physical exam findings of ILD can be related both to the lung disease and to associated systemic diseases.
- ILD can be primary (idiopathic) or secondary.
- Common secondary causes of ILD include chronic pneumonias, inhalation of dusts, systemic inflammatory/autoimmune disease, certain drugs, and radiation.
- Drug-induced ILD can often occur after use of chemotherapy drugs like bleomycin.
- Systemic diseases that cause ILD include sarcoidosis, lupus, rheumatoid arthritis, and systemic sclerosis.
- ILD is characterized by persistent inflammation with excessive fibrosis.
- The resultant restrictive lung disease reduces lung volumes and leads to hypoxemia and dyspnea over time.
- Physical findings include crackles on lung auscultation, cyanosis, and clubbed fingers.
- Chest imaging may show a bibasilar reticular pattern and honeycombing.
- Spirometry shows a restrictive pattern, with decreases in most lung volumes and capacities.
- Arterial blood gas shows hypoxemia and respiratory alkalosis.
- A bronchoalveolar lavage or lung biopsy can be performed in selected cases to make a more precise diagnosis and exclude other conditions.

Review Questions

1. **Which of the following radiologic findings is most likely to be found on the chest x-ray of a patient with interstitial lung disease?**

   A. A solitary pulmonary nodule

   B. An interstitial infiltrate in one lobe

   C. Hyperinflated lungs

   D. Multiple large pulmonary nodules

   E. Reticular infiltrates in multiple lobes

Explanation **(requires correct answer)**

2. **Which of the following pulmonary function test results is least likely in a patient with interstitial lung disease?**

   A. Decreased functional residual capacity

   B. Decreased forced vital capacity

C. Decreased FEV_1/FVC ratio

D. Decreased residual volume

E. Normal FEV_1/FVC ratio

Explanation (requires correct answer)

3. **A 61-year-old female patient presents with a persistent dry cough and shortness of breath that have been slowly worsening over the past 6 months. She reports no history of smoking. She was diagnosed with rheumatoid arthritis 15 years ago and treated successfully with a combination of methotrexate and sulfasalazine. Physical examination reveals crackles on lung auscultation and incipient clubbing of the fingers, and a decreased oxygen saturation on oximetry. Which of the following would be the best initial diagnostic study?**

A. Chest CT scan

B. Chest x-ray

C. Complete blood count

D. Serologic studies

E. Spirometry

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 Explain the use of the term interstitial lung disease (ILD).

02 Describe the clinical symptoms for patients with interstitial lung disease.

03 List the most common causes of interstitial lung disease, including medications.

04 Explain how interstitial lung disease affects pulmonary pathology and physiology.

05 Summarize the diagnostic approach to ILD.

Objective Confidence: 0/5